

Introduction to Geographic Information Systems (GIS)

Presented by: Brent Fallarcuna
GIS Specialist

Presentation Outline

This is an intensive short course for Geographic Information System that includes basic concept and applications of GIS.

**1. WHAT IS A GEOGRAPHIC
INFORMATION SYSTEM?**

**2. COMPONENTS OF GEOGRAPHIC
INFORMATION SYSTEMS**

3. MAP COMPONENTS

4. MAP PROJECTIONS

1

What is a Geographic Information System?

Understanding the basic concepts of GIS and its applications.

“A geographic information system (GIS) is a computer system for capturing, storing, checking, and displaying data related to positions on Earth’s surface. GIS can show many different kinds of data on one map, such as streets, buildings, and vegetation. This enables people to more easily see, analyze, and understand patterns and relationships”

“A GIS is a specific information system applied to geographic data and is mainly referred to as a system of hardware, software, personnel and procedures designed to support capture, management, manipulation, analysis, modeling and display of spatially referenced data for solving complex planning and management problems.” [Burroughs, 1986; NCGIA, 1990].

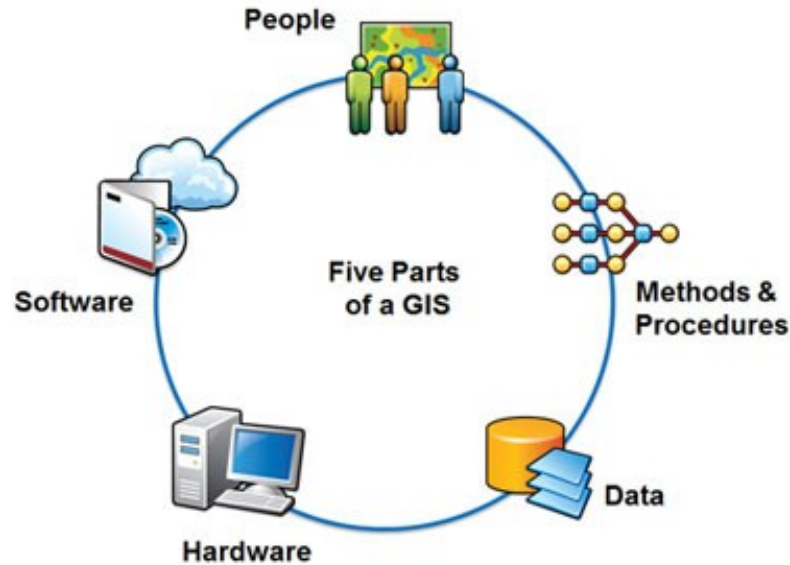
GIS Applications

- Forest Monitoring and Management
- Land Use Planning
- Health Care Monitoring
- Demography
- Disaster Risk Management
- **Agriculture**

2

Components of a GIS

Understanding the key elements of a GIS



GIS Components

A GIS is composed of the following: (1) Hardware, (2) Software, (3) Data, (4) People and (5) Methods.

COMPONENT: HARDWARE

Input Devices

- Keyboard
- Digitizing Table
- Scanner
- Mouse
- GPS units

Central Processing Unit

- External Disks
- CDs / DVDs

Output Devices

- Monitor
- Printer
- Plotter

COMPONENT: SOFTWARE

- Data Management System
- Supports data input and data manipulation
- Supports geographic query, analysis and visualization
- Geographic User Interface (GUI)



COMPONENT: DATA

- a. Data Sources
- b. Geographic Data Characteristics
- c. Data Types
- d. Data Structure

COMPONENT: DATA

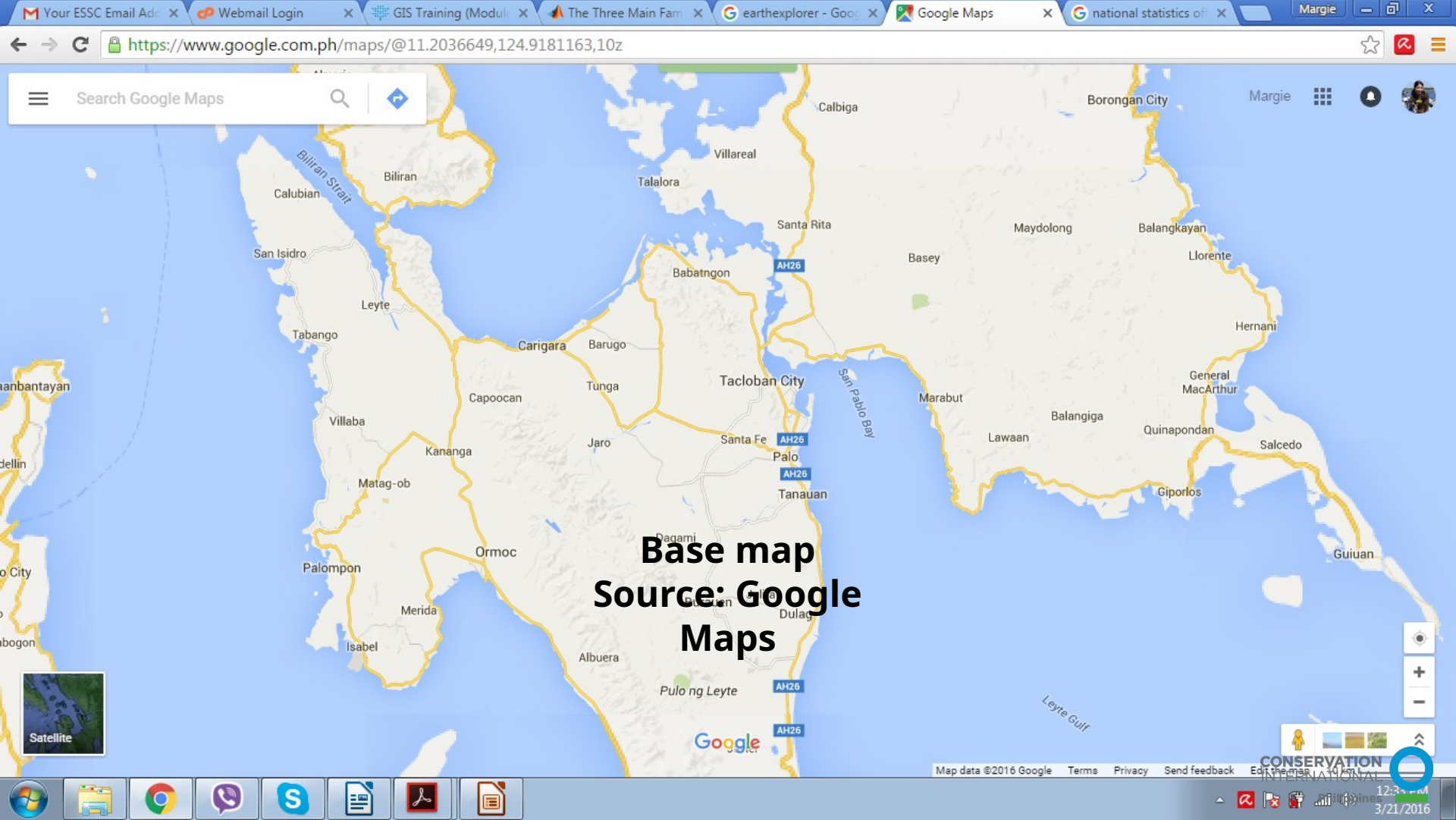
- a. Data Sources**
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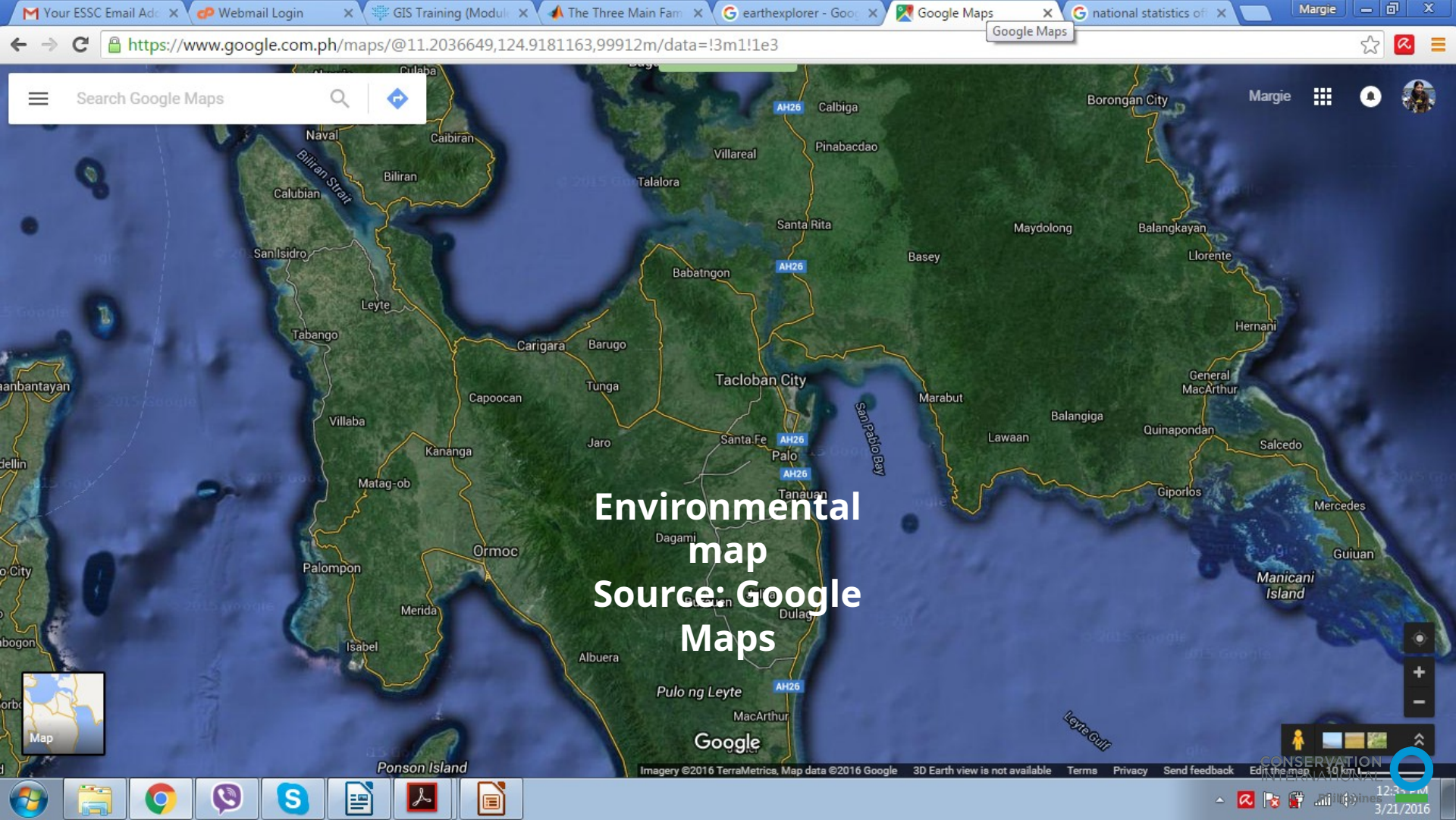
What types of data do GISs use?

Base Maps – includes streets, highways, administrative boundaries for census, postal and political areas. Also rivers, lakes, parks and landmarks

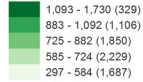
Environmental maps – includes data related to the environment, weather, climate, environmental risk, satellite imageries, topography and natural resources.

Socio-economic data – include data related to census/demography, health care, real estate, telecommunications, emergency preparedness, crime, businesses and transportation.

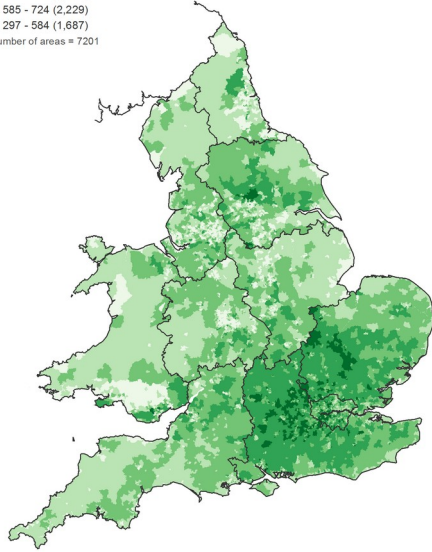




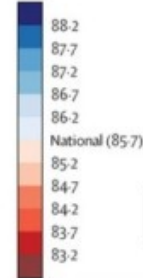
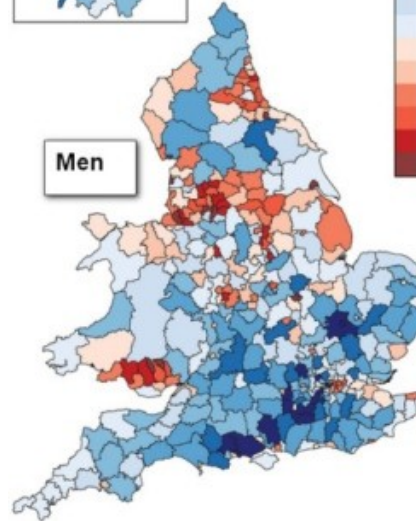
Total weekly income (£)



Total number of areas = 7201

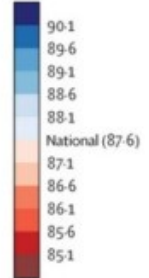
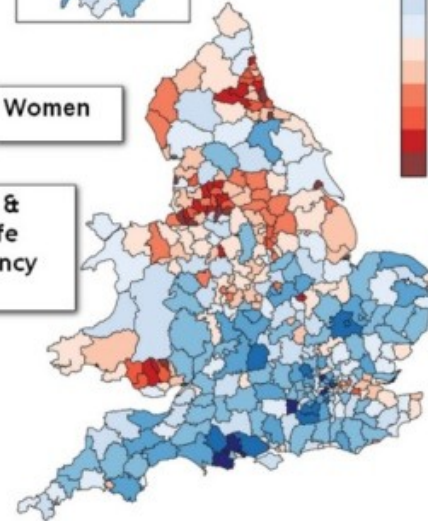


Men



Women

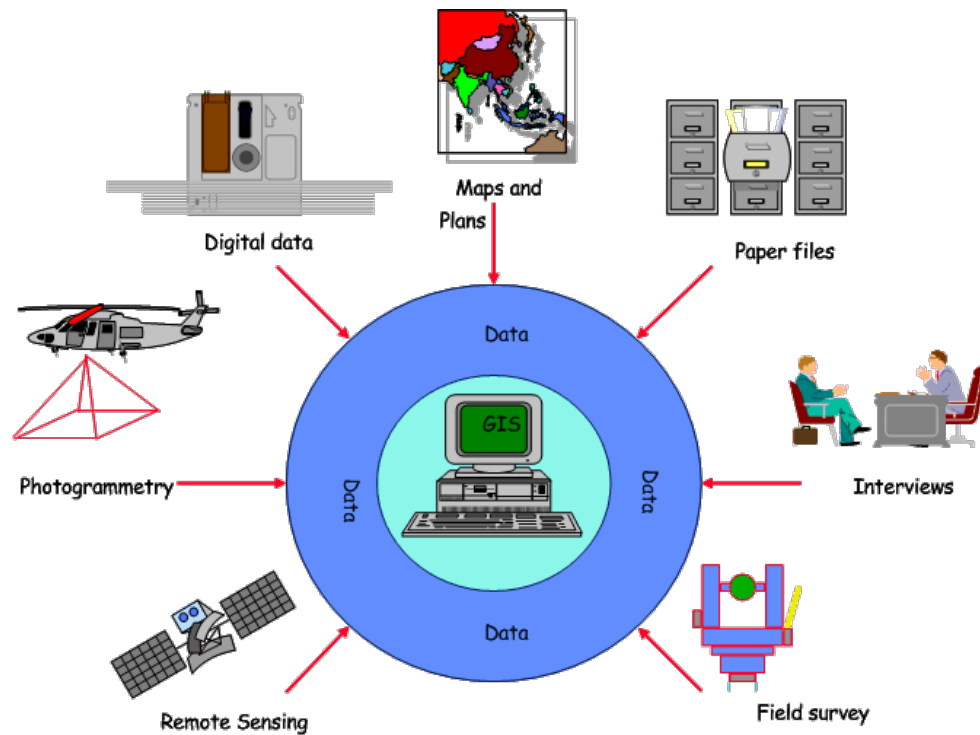
England & Wales life expectancy 2030



Socio-Economic and demographic data maps

Source:

<http://www.ons.gov.uk/employmentandlabourmarket/peopleinwork/earningsandworkinghours/bulletins/smallareamodelbasedincomeestimates/>



Data Sources

National Mapping and Resources Information Authority (NAMRIA)

Topographic Maps

1:2500,00 Scale

1:50,000 Scale

Ready Hazard Maps

Earthquake-induced Landslide

Ground Rapture

Ground Shaking

Liquefaction

Storm Surge

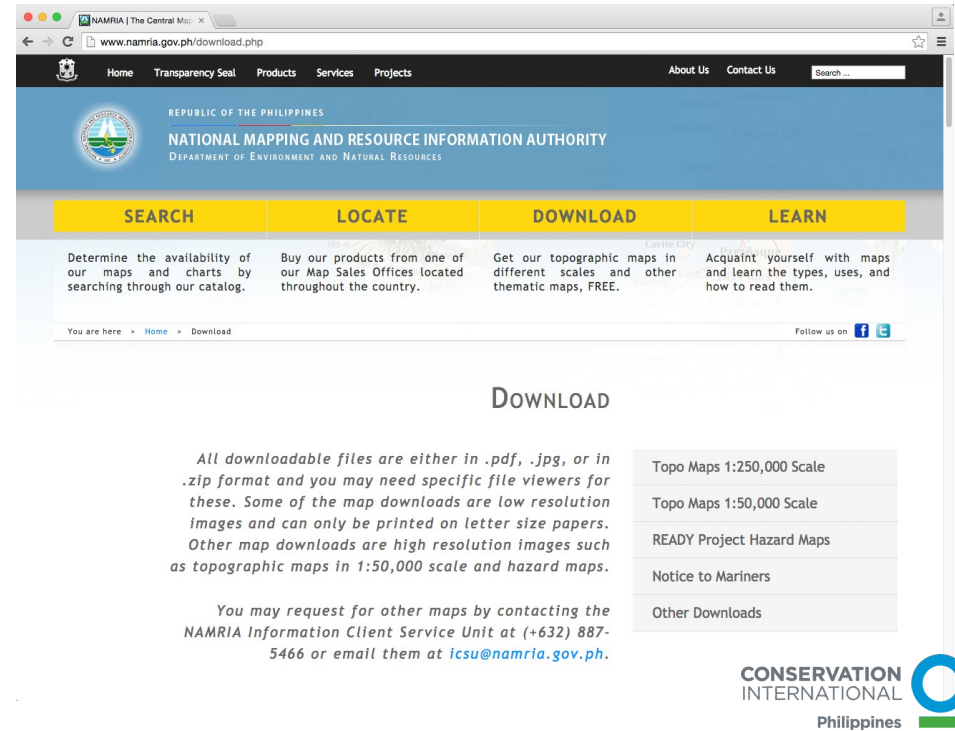
Tsunami

Rain-induces Landslide

Flood map (1:50,000 scale)

Flood map (1:10,000 scale)

Volcanic



The screenshot shows the NAMRIA website interface. At the top, there's a navigation bar with links: Home, Transparency Seal, Products, Services, Projects, About Us, and Contact Us. Below this is a header section with the NAMRIA logo and the text "REPUBLIC OF THE PHILIPPINES NATIONAL MAPPING AND RESOURCE INFORMATION AUTHORITY DEPARTMENT OF ENVIRONMENT AND NATURAL RESOURCES". A yellow navigation bar contains four main sections: SEARCH, LOCATE, DOWNLOAD, and LEARN. Each section has a brief description of its function. The "DOWNLOAD" section is highlighted, showing a list of downloadable files: Topo Maps 1:250,000 Scale, Topo Maps 1:50,000 Scale, READY Project Hazard Maps, Notice to Mariners, and Other Downloads. A note mentions that files are in .pdf, .jpg, or .zip format. Contact information for the NAMRIA Information Client Service Unit is provided at the bottom.

DOWNLOAD

All downloadable files are either in .pdf, .jpg, or in .zip format and you may need specific file viewers for these. Some of the map downloads are low resolution images and can only be printed on letter size papers. Other map downloads are high resolution images such as topographic maps in 1:50,000 scale and hazard maps.

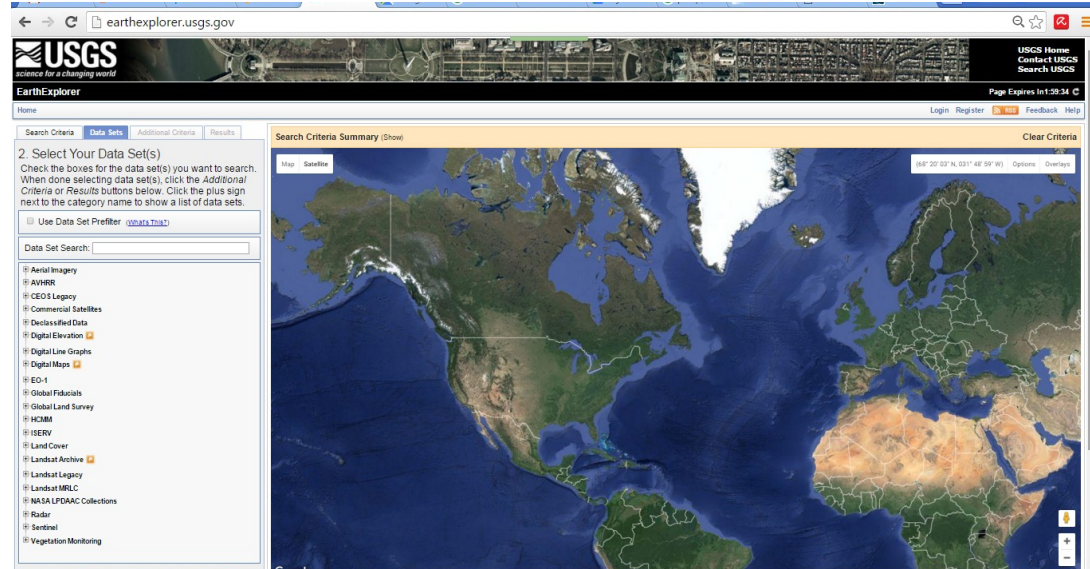
You may request for other maps by contacting the NAMRIA Information Client Service Unit at (+632) 887-5466 or email them at icsu@namria.gov.ph.

- Topo Maps 1:250,000 Scale
- Topo Maps 1:50,000 Scale
- READY Project Hazard Maps
- Notice to Mariners
- Other Downloads

CONSERVATION INTERNATIONAL
Philippines

Other data sources

- Surveying Companies
- Google Earth
- Google Maps
- Bing Maps
- Free Data:
 - Earth Explorer
 - FreeGIS (freegis.org)
 - PhilGIS (philgis.org)
 - OpenStreetMap (openstreetmap.org)



COMPONENT: DATA

a. Data Sources

**b. Geographic Data
Characteristics**

c. Data Types

d. Data Structure

Data Characteristics

- Spatial components
 - Location
 - Attribute
 - Spatial Relationship
 - Time
- Geographic data requires reference data
- GIS includes operations which supports spatial analysis (e.g. neighborhood analysis)

COMPONENT: DATA

- a. Data Sources
- b. Geographic Data Characteristics
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Data Types

Spatial Data

Objects that have a geographic reference.

These can be translated into objects such as points, lines, area and grids.

These are used to represent objects into maps

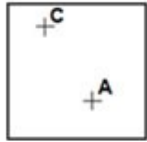
Example: a parcel of land

Attribute Data (Non-spatial / Aspatial Data)

These are the descriptive information of specific spatial objects. Often it have no direct information about the spatial location but can be link to the spatial object.

Example: the name of the owner of a parcel land

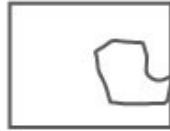
Vector Data Model



(a)Point



(b)Line



(c)Polygon

Spatial Data

Points – can represent posts, a stoplight or a manhole

Lines – can represent roads

Polygon – can represent an area or property

Exercise

Identify the following if Spatial

or Aspatial:

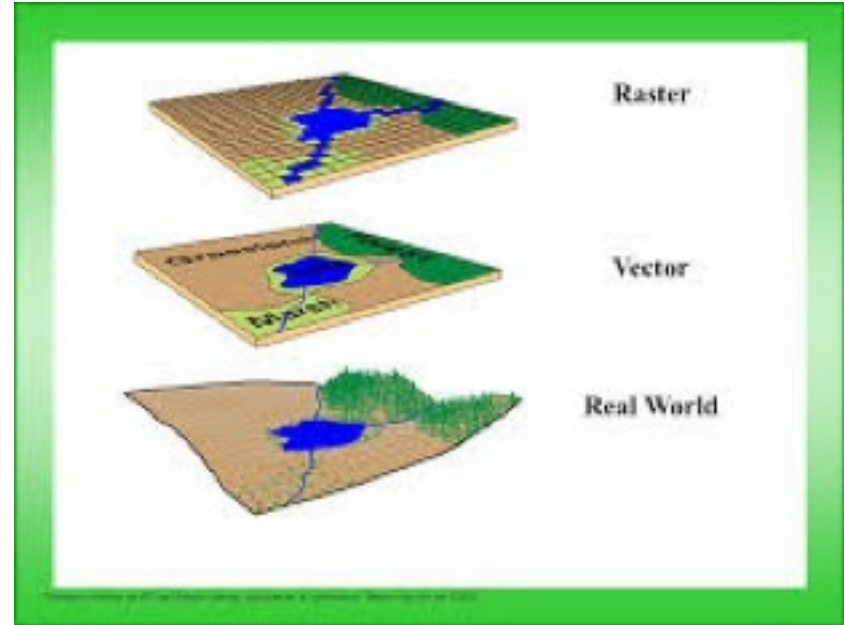
- Building
- Color of the Building
- Name of the Building
- Samar
- Leyte
- Population of Samar

COMPONENT: DATA

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Data Structures

1. Raster – represented as cells
2. Vector – represented as points, lines and polygons.



3

Mapping Essentials

Understanding the graphical representation of the features on a plane surface that appear on the curved surface of the earth.

What is a Map?

- A representation, normally scaled, of features on the surface of the earth
- A graphical depiction of all or part of a graphical realm in which the real-world features have been replaced by symbols in their correct spatial location at a reduced scale.

Uses of Maps

- Economic or Physical Framework Planning
- Population Health Monitoring
- Social Welfare and Development
- Land and Public Works
- Environmental Planning and Management
- Agricultural Management
- and many, many more!

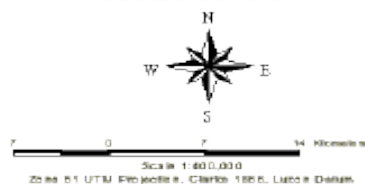
Basic Map Contents (Elements)

- **Primary Contents** – includes the subject of the map. e.g. land use, geology, topography
- **Secondary Contents** – includes graticules and values of longitude and latitude around the map sheets. e.g. administrative boundaries, roads, grids
- **Supportive Contents** – marginal information such as scale, legend, location, diagram, projection, north orientation, explanation text.

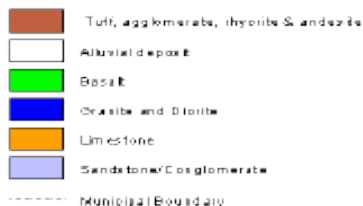
Supportive Content

UPPER MAGAT WATERSHED NUEVA VIZCAYA, PHILIPPINES

GEOLOGY MAP



Legend:



Source of Information:
Geologic Map, DENR/ENRDO, Nueva Vizcaya
Scale: 1:25,000

Source of Digital Data:
NAMRIA, Digital Data, 1995
Note: Administrative boundaries are approximate.

LAND COVER AND LAND USE CHANGE CASE STUDY - PHASE II

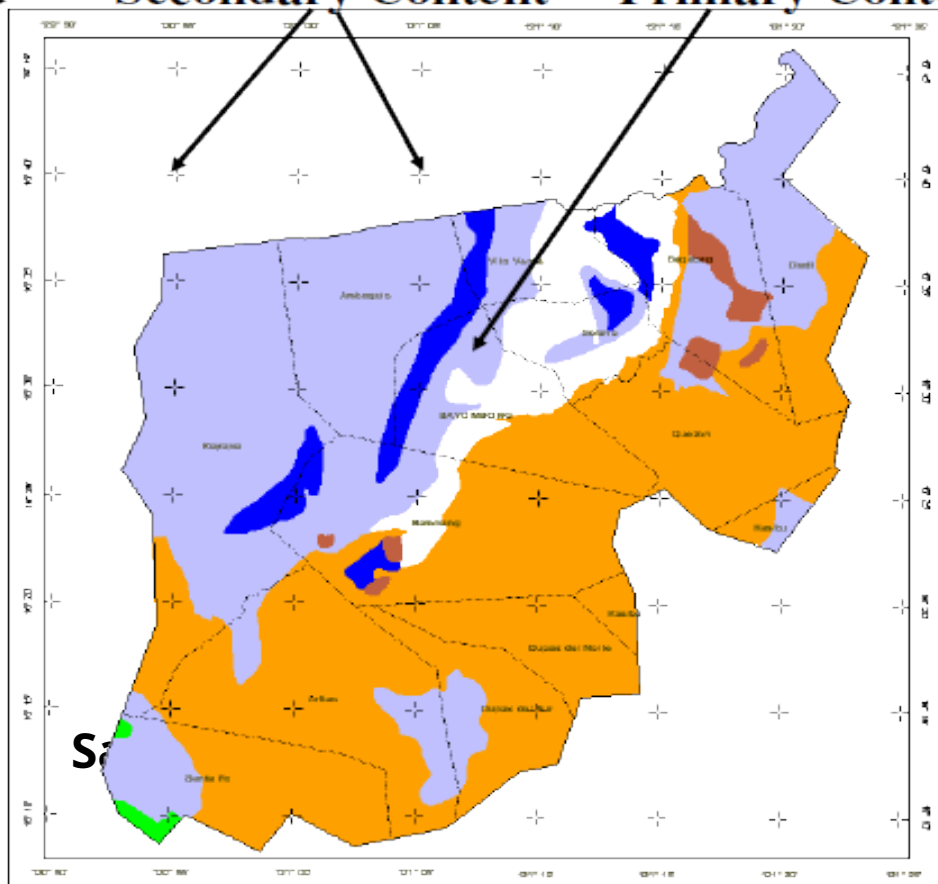
Produced By:

National Mapping and Resource Information Authority
Philippine Meteorological, Climatological & Astronomical
Service, Alabang, Muntinlupa
National Science and Technology Development Authority



Secondary Content

Primary Content



Map Components

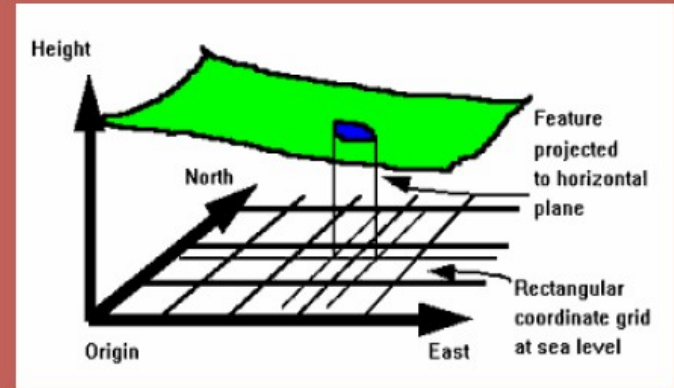
1. **Map title** – defines the map's information that includes geographical area and feature that is being represented - .
2. **Legend / Key** – defines the different symbols in the map means.
3. **North Arrow / Compass** – defines the ordinal orientation of the map.
4. **Scale** – defines the ratio of the maps representation to the real world objects.
5. **Grid lines** – show the latitude and longitude. It is important to know the projection that was used.
6. **Explanatory Text** – added notes that are important for the map interpretation
7. **Data Souce Text** – includes data source, when was the data processed, date when the map was published, and name of the institution that published the map.

Reference Model

Flat Earth Representation

* for a larger scale of the earth's surface, the flat earth can be distorted.

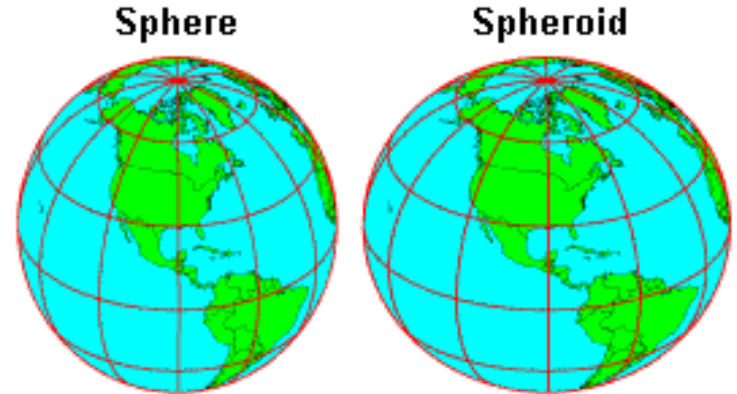
"Flat Earth" Model



Reference Model: Spheroid

Curved Earth Model

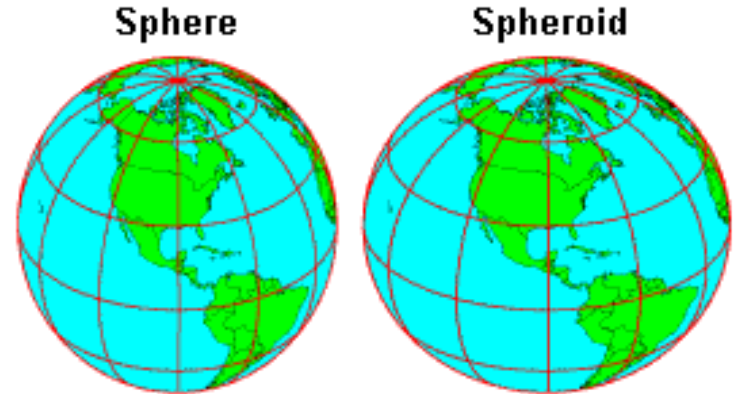
A spheroid or ellipsoid is a quadratic surface obtained by rotating an ellipse about one of its principal axes. This model has the advantage of accommodating the bulge at the equator while remaining relatively mathematically simple.



Reference Model: Spheroid

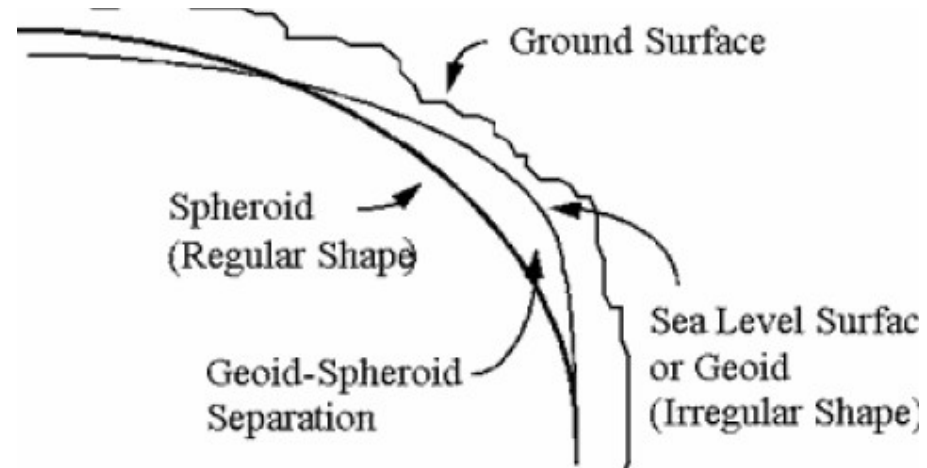
Curved Earth Model

The irregularity of the shape of the earth's surface is due to different compositions of the earth surface, e.g. variation of density of rock and its distribution on earth surface. This causes the irregularity in the surface of the sea level which is the shape of the gravity field.



Reference Model: Spheroid

Sea level is widely used as the reference surface for the measurement of height. The contours on a map will usually show height above sea level (i.e. MASL). However, their position will be mapped on the spheroid.

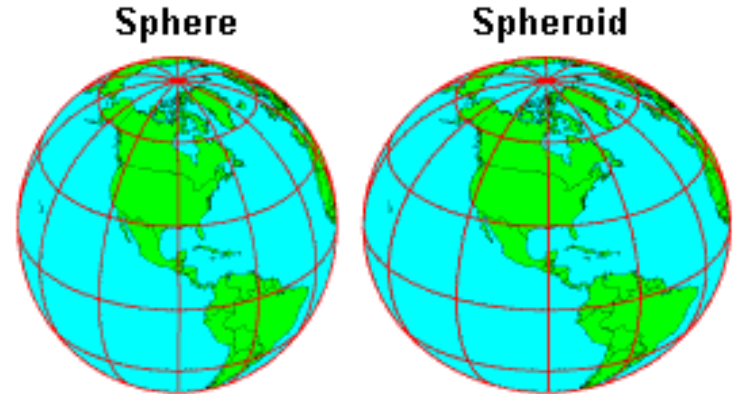


Reference Model: Spheroid

Curved Earth Model is also referred to as **Geodetic Datum**.

2 Categories of Geodetic Datums:

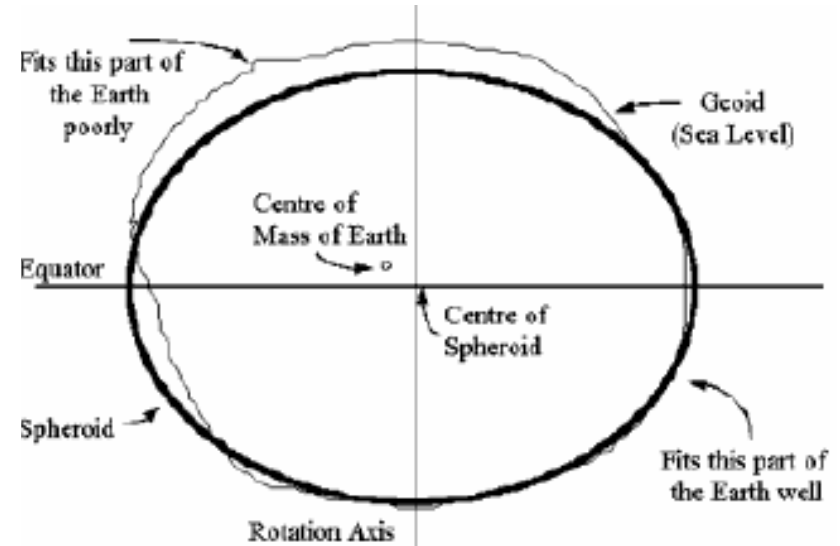
- a. Local Geodetic Datums
- b. Geocentric Datums



Reference Model: Spheroid

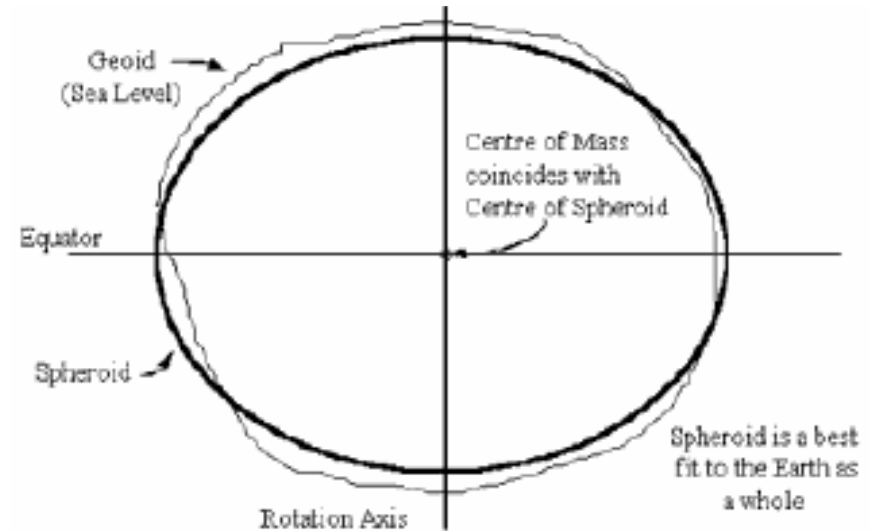
Local Geodetic Datum

A datum which best approximates the size and shape of a particular part of the earth's sea-level surface. Invariably, the centre of its spheroid will not coincide with the Earth's centre of mass.



Reference Model: Spheroid Geocentric Datum

One which best approximates the size and shape of the Earth as a whole. The centre of its spheroid coincides with the Earth's centre of mass. Not a good approximation of any particular part of the Earth. Useful in projects or undertakings which have global application. Used by GPS.

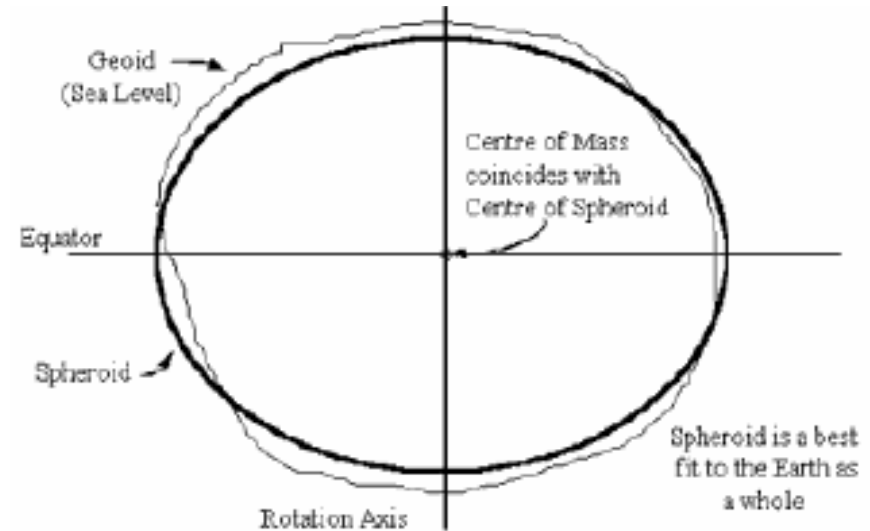


Reference Model: Spheroid

Geocentric Datum

There are 2 coordinate systems that can be associated with geodetic datums:

1. **Geodetic coordinate system** – naturally associated with the spheroid and allows positions on the earth's surface to be described in terms of latitude, longitude and spheroidal height
2. **Cartesian coordinate system** uses X, Y and Z axes



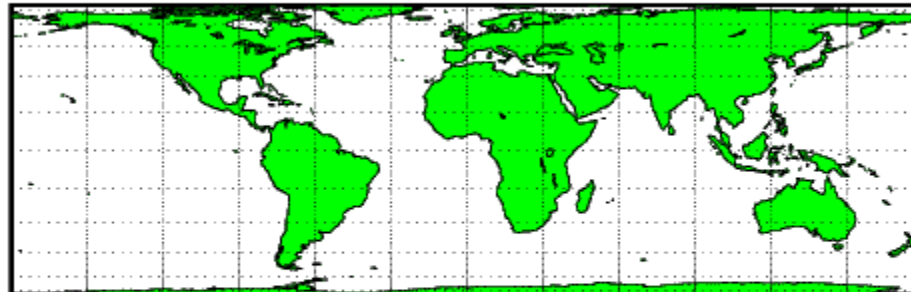
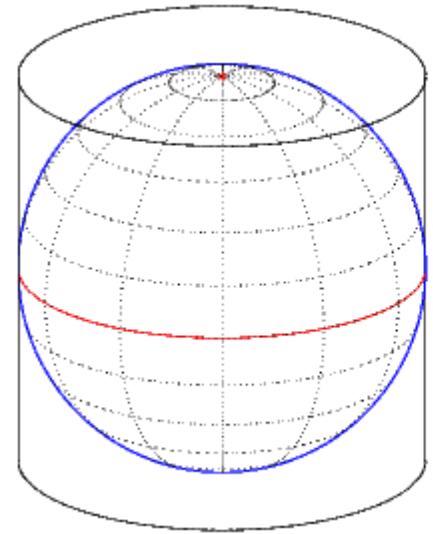
3 Map Projections

A map projection is a systematic transformation of the latitudes and longitudes of locations on the surface of a sphere or an ellipsoid into locations on a flat plane

Types of Map Projections

1. Cylindrical - true at the equator and distortion increases toward the poles.

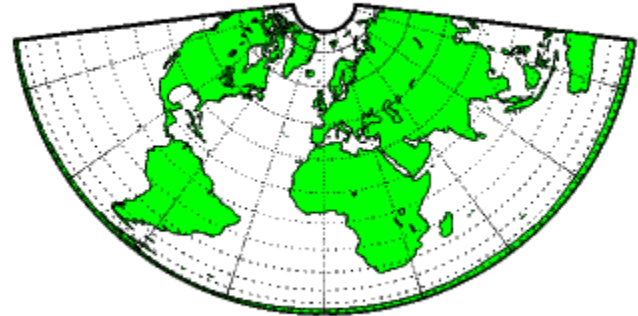
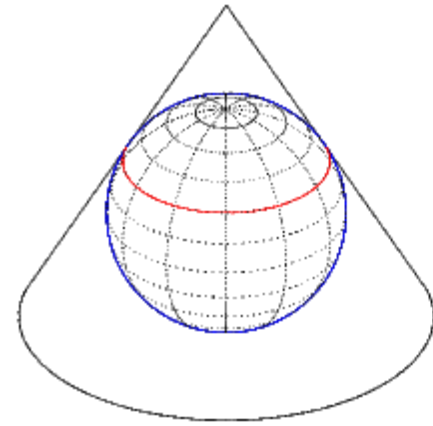
- *countries in the tropics are best suited for a cylindrical projection*



Types of Map Projections

2. Conic - true along some parallel somewhere between the equator and a pole and distortion increases away from this standard.

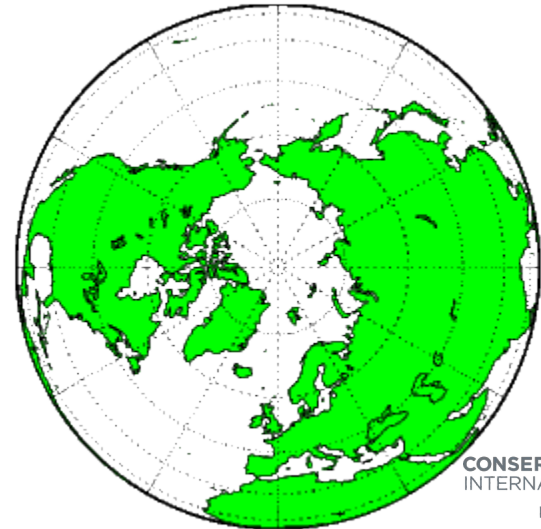
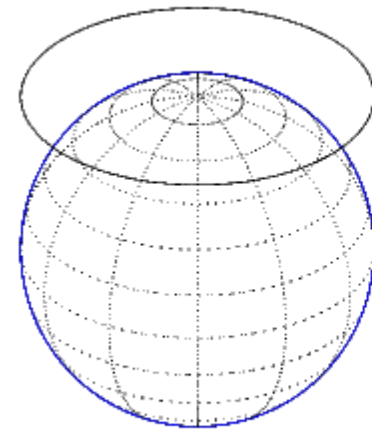
- *countries in the temperate zone are best suited for a conical projection*



Types of Map Projections

3. Planar - true only at their center point, but generally distortion is worst at the edge of the map.

- *polar areas are best suited for an azimuthal projection.*



Sample Projections

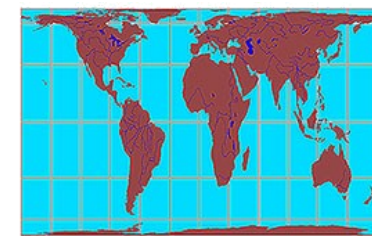
Projections for the Philippines:

a. WGS 84 / UTM zones 50N, 51N, and 52N

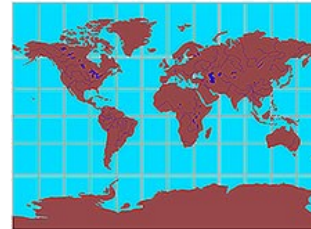
b. Luzon Datum PRS92



Mercator Projection



Gall-Peters Projection



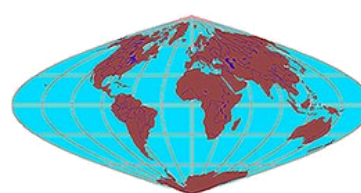
Miller Cylindrical Projection



Mollweide Projection



Goode's Homolosine Equal-area Projection



Sinusoidal Equal-Area Projection



Robinson Projection

Salamat!